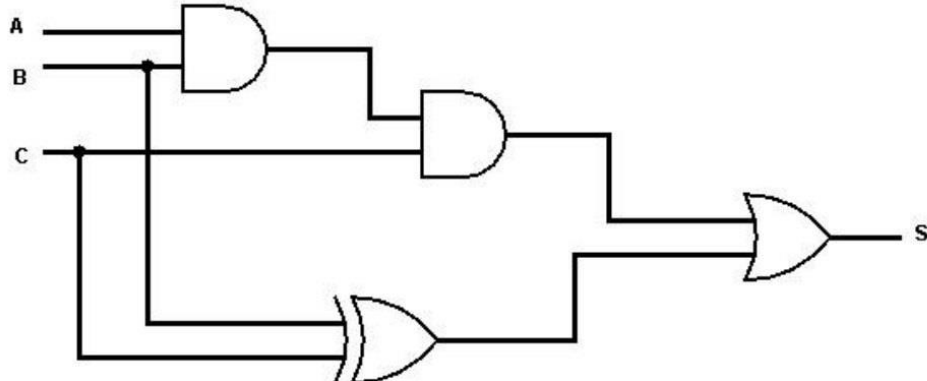


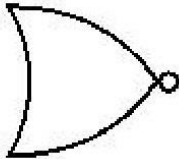
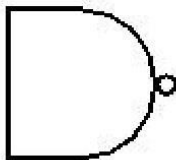
8	1 mark per correct row	3																								
<table><tr><th>Statement</th><th>AND</th><th>NAND</th><th>NOR</th><th>XOR</th><th>OR</th></tr><tr><td>The output is 1 only when both inputs are 1</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>The output is 1 only when both inputs are different</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>The output is 1 only when both inputs are 0</td><td></td><td></td><td>✓</td><td></td><td></td></tr></table>			Statement	AND	NAND	NOR	XOR	OR	The output is 1 only when both inputs are 1	✓					The output is 1 only when both inputs are different				✓		The output is 1 only when both inputs are 0			✓		
Statement	AND	NAND	NOR	XOR	OR																					
The output is 1 only when both inputs are 1	✓																									
The output is 1 only when both inputs are different				✓																						
The output is 1 only when both inputs are 0			✓																							

3(a)	1 mark for each correct gate, with correct inputs 	4																																													
3(b)	1 mark for each half (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>S</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	S	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C	Working space	S																																											
0	0	0		0																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		0																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

8	1 mark per correct row	3																								
<table><tr><th>Statement</th><th>AND</th><th>NAND</th><th>NOR</th><th>XOR</th><th>OR</th></tr><tr><td>The output is 1 only when both inputs are 1</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>The output is 1 only when both inputs are different</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>The output is 1 only when both inputs are 0</td><td></td><td></td><td>✓</td><td></td><td></td></tr></table>			Statement	AND	NAND	NOR	XOR	OR	The output is 1 only when both inputs are 1	✓					The output is 1 only when both inputs are different				✓		The output is 1 only when both inputs are 0			✓		
Statement	AND	NAND	NOR	XOR	OR																					
The output is 1 only when both inputs are 1	✓																									
The output is 1 only when both inputs are different				✓																						
The output is 1 only when both inputs are 0			✓																							

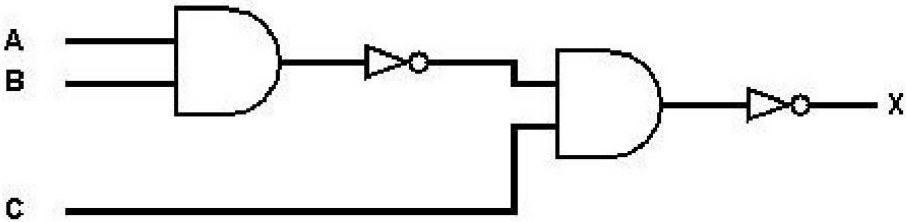
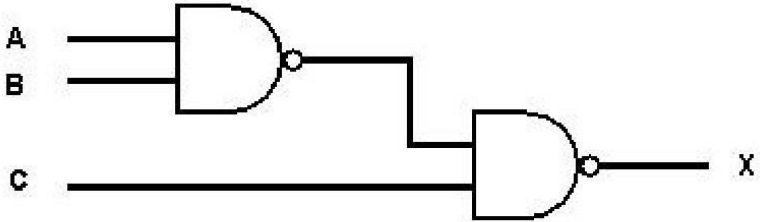
3(a)	1 mark per bullet point • A AND B ... • ... XOR C ... • ... OR NOT B ((A AND B) XOR C) OR NOT B	3
------	--	----------

Answer 4 (continued)

3(b)	1 mark for each set of 4 rows (shaded) <table><tr><td>A</td><td>B</td><td>C</td><td>Working space</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		0																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		0																																											
3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> NAND  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	OUTPUT	0	0	1	0	1	0	1	0	0	1	1	0	A	B	OUTPUT	0	0	1	0	1	1	1	0	1	1	1	0	3															
A	B	OUTPUT																																													
0	0	1																																													
0	1	0																																													
1	0	0																																													
1	1	0																																													
A	B	OUTPUT																																													
0	0	1																																													
0	1	1																																													
1	0	1																																													
1	1	0																																													

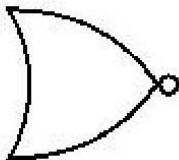
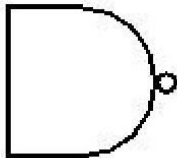
2(a)	1 mark for each completely correct truth table NOR <table border="1" data-bbox="297 291 644 636"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND <table border="1" data-bbox="297 707 644 1052"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0	A	B	Output	0	0	1	0	1	1	1	0	1	1	1	0	2
A	B	Output																														
0	0	1																														
0	1	0																														
1	0	0																														
1	1	0																														
A	B	Output																														
0	0	1																														
0	1	1																														
1	0	1																														
1	1	0																														

Answer 5 (continued)

2(b)	1 mark for each correct bullet point • NOT (A AND B) // A NAND B • NOT the result AND C // the result NAND C  The diagram shows a logic circuit with three inputs: A, B, and C. Inputs A and B are connected to an AND gate. The output of this AND gate is connected to a NOT gate (inverter). The output of the NOT gate is connected to one input of a second AND gate. Input C is connected to the other input of the second AND gate. The output of the second AND gate is connected to a final NOT gate, which produces the output X. OR  The diagram shows an alternative logic circuit for the same function. Inputs A and B are connected to an AND gate. The output of this AND gate is connected to one input of a second AND gate. Input C is connected to the other input of the second AND gate. The output of the second AND gate is connected to a final NOT gate, which produces the output X.	2
------	---	---

3(a)	1 mark per bullet point • A AND B ... • ... XOR C ... • ... OR NOT B ((A AND B) XOR C) OR NOT B	3
------	--	----------

Answer 6 (continued)

3(b)	1 mark for each set of 4 rows (shaded) <table><tr><td>A</td><td>B</td><td>C</td><td>Working space</td><td>X</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		0																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		0																																											
3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> NAND  <table><tr><td>A</td><td>B</td><td>OUTPUT</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	OUTPUT	0	0	1	0	1	0	1	0	0	1	1	0	A	B	OUTPUT	0	0	1	0	1	1	1	0	1	1	1	0	3															
A	B	OUTPUT																																													
0	0	1																																													
0	1	0																																													
1	0	0																																													
1	1	0																																													
A	B	OUTPUT																																													
0	0	1																																													
0	1	1																																													
1	0	1																																													
1	1	0																																													

2(c)(i)	1 mark for each correct row. <table border="1"> <thead> <tr> <th>Gate</th><th>Input 1</th><th>Input 2</th></tr> </thead> <tbody> <tr> <td>AND</td><td>1</td><td>1</td></tr> <tr> <td>NAND</td><td>0 0 1</td><td>0 1 0</td></tr> <tr> <td>XOR</td><td>0 1</td><td>1 0</td></tr> <tr> <td>NOR</td><td>0</td><td>0</td></tr> </tbody> </table>	Gate	Input 1	Input 2	AND	1	1	NAND	0 0 1	0 1 0	XOR	0 1	1 0	NOR	0	0	4
Gate	Input 1	Input 2															
AND	1	1															
NAND	0 0 1	0 1 0															
XOR	0 1	1 0															
NOR	0	0															
2(c)(ii)	1 mark per bullet point • A NAND B • B XOR C • OR (A NAND B) OR (B XOR C)	3															

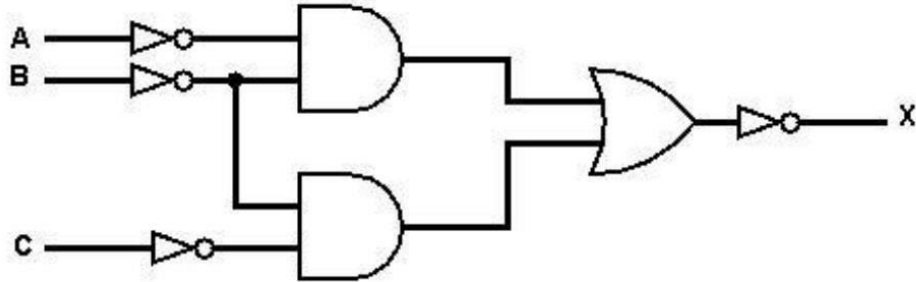
7	1 mark for first 4 rows, 1 mark for second 4 rows (shaded)	2																																													
<table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>			A	B	C	Working space	X	0	0	0		0	0	0	1		0	0	1	0		1	0	1	1		0	1	0	0		1	1	0	1		0	1	1	0		0	1	1	1		0
A	B	C	Working space	X																																											
0	0	0		0																																											
0	0	1		0																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		1																																											
1	0	1		0																																											
1	1	0		0																																											
1	1	1		0																																											

7(a)	1 mark for each section • A AND B • NOT C AND B // B AND NOT C • XOR (with remainder correct and bracketed and nothing extra) X = (A AND B) XOR (NOT C AND B)	3																																													
7(b)	1 mark for first 4 rows, 1 mark for second 4 rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	2
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		0																																											

3(a)

1 mark for each bullet point:

- NOT A AND NOT B **and** NOT B AND NOT C // A NOR B **and** B NOR C
- final OR and NOT gates (with correct inputs) // NOR gate (with correct inputs)



2

Answer 10 (continued)

3(b)

1 mark for each set of rows as highlighted:

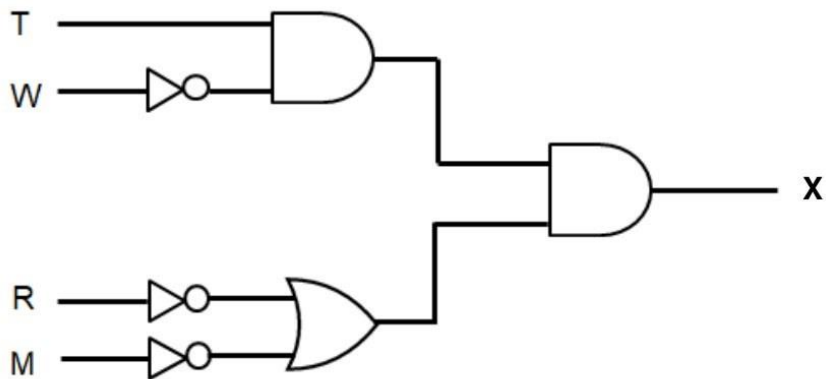
2

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

3(a)

1 mark for T AND NOT W
1 mark for NOT R OR NOT M
1 mark for final AND

3



3(b)

1 mark for each set of rows as highlighted:

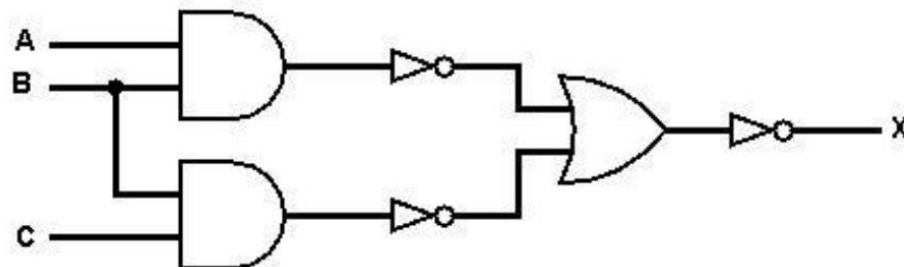
2

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

5(a)

1 mark for each bullet point:

- NOT (A AND B)
- NOT (B AND C)
- NOT(NOT(A AND B) OR NOT(B AND C))



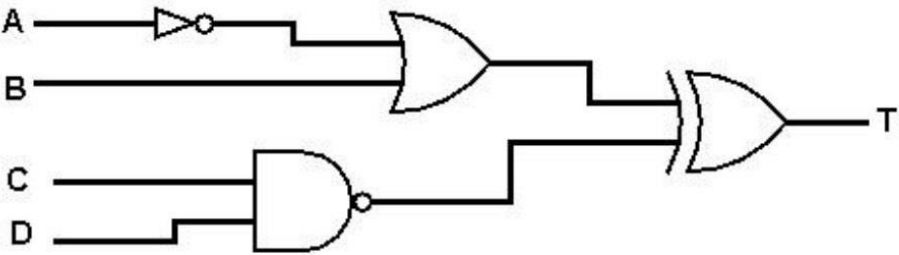
3

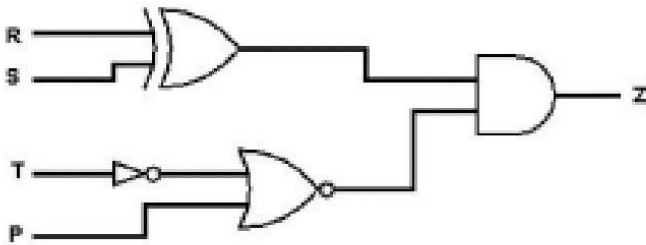
5(b)

1 mark for each set of highlighted rows.

P	Q	R	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

2

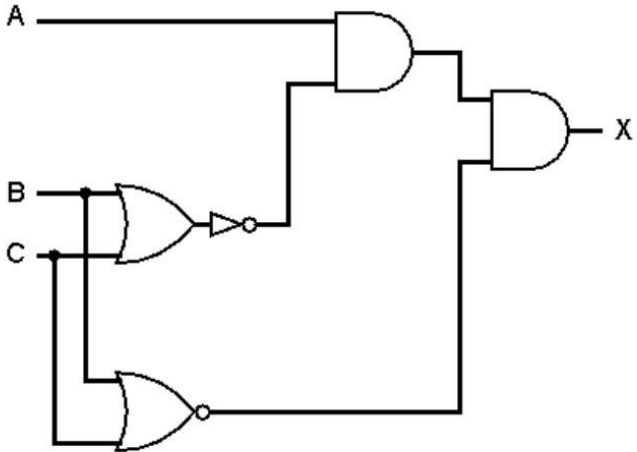
5(a)	1 mark for 2 gates 2 marks for all 4 gates 	2
5(b)	1 mark each NAND 0 is only output when both inputs are 1 // 1 is only output when none, or (either) one of the inputs is 1 NOR 1 is only output when both inputs are 0 // 0 is only output when (either) one or both inputs are 1	2

6(a)	1 mark for correct XOR and AND gates, with correct inputs 1 mark for correct NOT and NOR gates with correct inputs 	2																																													
6(b)	1 mark for first 4 rows 1 mark for last 4 rows <table><thead><tr><th>P</th><th>Q</th><th>R</th><th>Working space</th><th>Z</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></tbody></table>	P	Q	R	Working space	Z	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		0	1	1	0		1	1	1	1		1	2
P	Q	R	Working space	Z																																											
0	0	0		0																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		0																																											
1	1	0		1																																											
1	1	1		1																																											

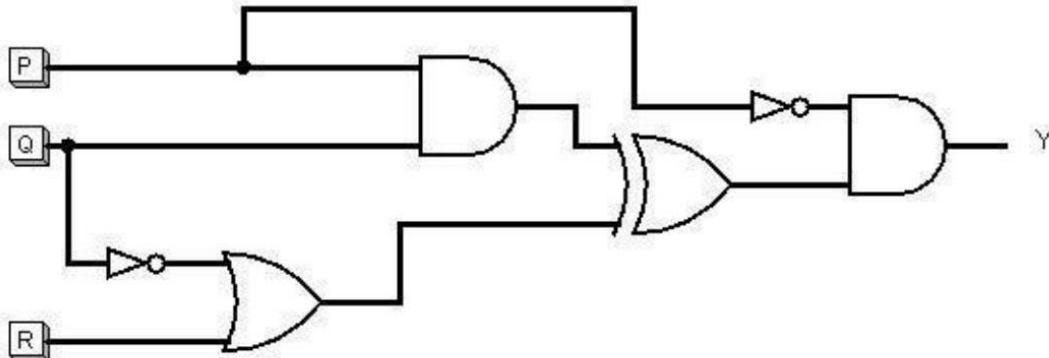
1(a)	NOT B	1																																				
1(b)	1 mark for first 4 rows correct; 1 mark for second 4 rows correct <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	0	1	1	1	1	2
A	B	C	X																																			
0	0	0	1																																			
0	0	1	0																																			
0	1	0	1																																			
0	1	1	0																																			
1	0	0	1																																			
1	0	1	0																																			
1	1	0	0																																			
1	1	1	1																																			

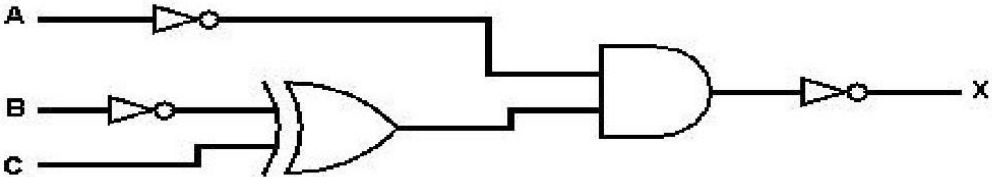
4(a)	1 mark for each set of 4 rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1	1	2
A	B	C	X																																			
0	0	0	1																																			
0	0	1	0																																			
0	1	0	0																																			
0	1	1	0																																			
1	0	0	1																																			
1	0	1	0																																			
1	1	0	1																																			
1	1	1	1																																			

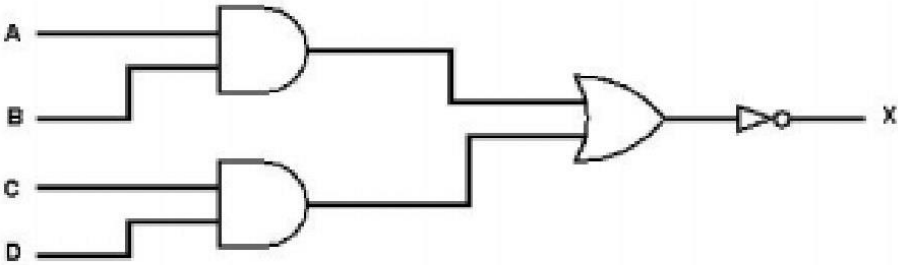
Answer 16 (continued)

4(b)	1 mark for both AND gates 1 mark for NOT gate <u>and</u> OR gate <u>and</u> NOR gate  <pre>graph LR; A --- AND1[AND]; B --- OR[OR]; C --- OR; C --- NOR[NOR]; OR --> NOT[NOT]; AND1 --> AND2[AND]; NOT --> AND2; NOR --> AND2; AND2 --> X[X];</pre>	2
------	---	----------

4(a)	1 mark for each bullet point • A NAND B • NOT(B XOR C) • Final NAND X = A NAND B NAND (NOT(B XOR C))	3																																				
4(b)	1 mark for each set of rows (shaded) <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	1	1	1	0	1	1	1	1	1	2
A	B	C	X																																			
0	0	0	0																																			
0	0	1	0																																			
0	1	0	1																																			
0	1	1	0																																			
1	0	0	1																																			
1	0	1	1																																			
1	1	0	1																																			
1	1	1	1																																			

4(a)	1 mark for each set of highlighted rows <table><tr><th>P</th><th>Q</th><th>R</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	P	Q	R	Y	0	0	0	1	0	0	1	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0	2
P	Q	R	Y																																			
0	0	0	1																																			
0	0	1	1																																			
0	1	0	0																																			
0	1	1	1																																			
1	0	0	0																																			
1	0	1	0																																			
1	1	0	0																																			
1	1	1	0																																			
4(b)	1 mark for P AND Q and NOT Q OR R 1 mark for NOT gate and AND gate and XOR gate 	2																																				

1(a)	1 mark for: $(A \text{ XOR } B) \text{ NOR } C$	1
1(b)	1 mark for NOT B XOR C 1 mark for NOT A and final AND plus NOT 	2

1(a)	1 mark for each correct answer: <i>NAND</i> - The output is 0 when both inputs are 1, otherwise the output is 1 <i>NOR</i> - The output is 1 when both inputs are 0, otherwise the output is 0 <i>XOR</i> - The output is 1 when one of the inputs is 1 and the other input is 0, otherwise the output is 0 <i>OR</i> - The output is 0 when both inputs are 0, otherwise the output is 1	4
1(b)	1 mark for both AND gates with correct inputs 1 mark for correct OR and NOT gates with correct inputs and no superfluous gates:  <pre> graph LR A --- AND1[AND] B --- AND1 C --- AND2[AND] D --- AND2 AND1 --- OR[OR] AND2 --- OR OR --- NOT[NOT] NOT --- X </pre>	2

6

1 mark for each correct line:

3

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Logic expression

NOT (A XOR B) AND C

(A OR C) AND NOT B

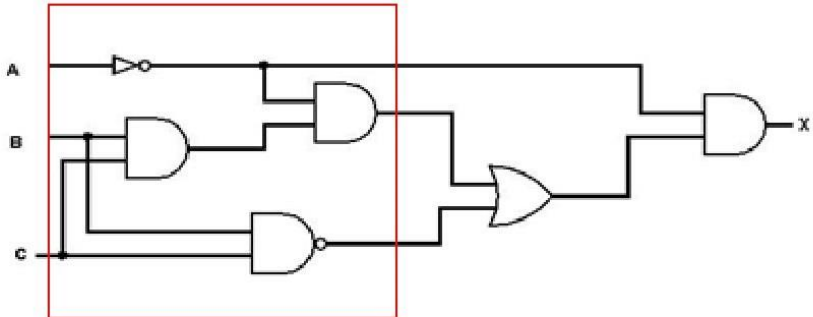
A NAND B NAND C

(A NAND B) OR C

NOT (A AND B AND C)

5(b)	1 mark for each correct expression X = E AND A AND C Y = E AND (B OR C OR D)	2
------	--	----------

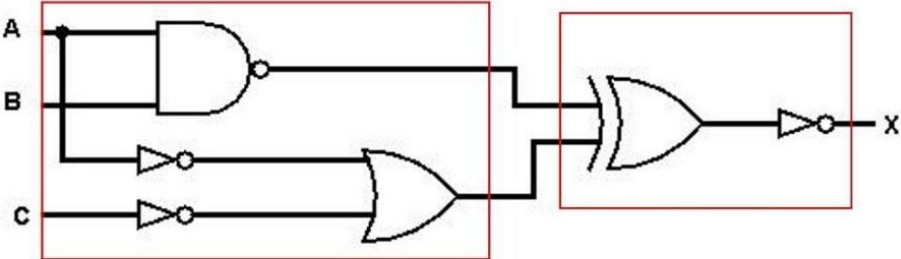
1(a)	1 mark for each shaded part. <table><thead><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></tbody></table>	A	B	C	X	0	0	0	1	0	0	1	1	0	1	0	1	0	1	1	1	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1	1	2
A	B	C	X																																			
0	0	0	1																																			
0	0	1	1																																			
0	1	0	1																																			
0	1	1	1																																			
1	0	0	1																																			
1	0	1	0																																			
1	1	0	1																																			
1	1	1	1																																			
1(b)	1 mark for (P XOR Q) and (Q OR NOT R) 1 mark for second XOR gate and final NAND gate with appropriate inputs	2																																				

1(a)	1 mark for: Output is only 1 if one input is 1 and the other is 0 // Output is only 1 if both inputs are different // Output is only 0 if both inputs are the same	1
1(b)	1 mark each: • $((A \text{ NOR } B) \text{ AND } (A \text{ OR } B))$ • $\text{NAND } (B \text{ AND } C)$ $X = ((A \text{ NOR } B) \text{ AND } (A \text{ OR } B)) \text{ NAND } (B \text{ AND } C)$	2
1(c)	1 mark for NOT gate, 2 correct AND gates and the NAND gate 1 mark for OR gate and the final AND gate 	2

1(a)	1 mark each: X = Q AND (NOT P OR R) Y = S XOR (Q NOR R)	2																																													
1(b)	1 mark for first 4 rows, 1 mark for second 4 rows <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		0	1	0	1		0	1	1	0		1	1	1	1		1	2
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		0																																											
0	1	1		1																																											
1	0	0		0																																											
1	0	1		0																																											
1	1	0		1																																											
1	1	1		1																																											

4(a)	1 mark each: • $X = (R \text{ NOR } S) \text{ XOR } T$ • $Y = (T \text{ AND } U) \text{ OR } U$	2																																													
4(b)	1 mark for first four rows, 1 mark for second four rows <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		0	1	1	0		0	1	1	1		1	2
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		0																																											
1	1	0		0																																											
1	1	1		1																																											

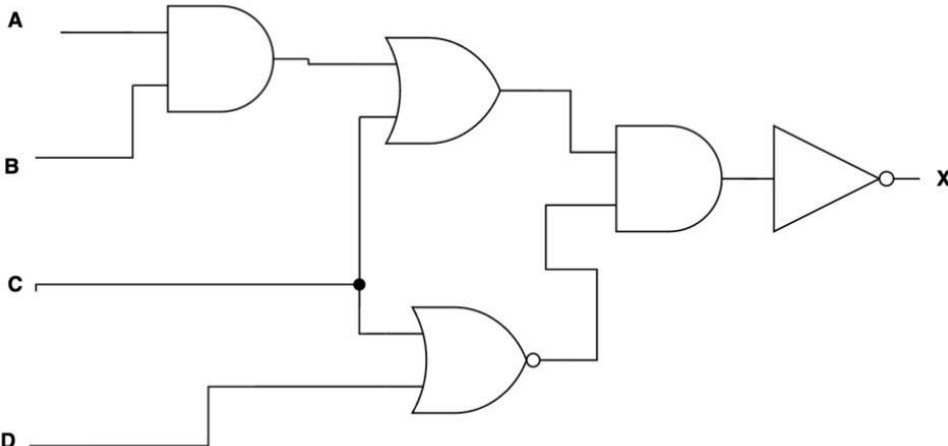
4(a)	1 mark for each correct statement $X = B \text{ NOR } (A \text{ NOR } C)$ $Y = (B \text{ AND } C) \text{ XOR NOT } D$	2																																													
4(b)	1 mark for 1st 4 rows 1 mark for 2nd 4 rows <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		0	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		1																																											
0	1	1		0																																											
1	0	0		0																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											

3(a)	One mark per bullet point, max 2 marks • (A NAND B) and (NOT A OR NOT C) • XOR and NOT 	2
3(b)	1 mark per bullet point, max 2 marks • One correct term: (R AND S AND NOT T) // (NOT R AND NOT S AND T) • Second correct term plus the OR in correct place Answer: Q = (R AND S AND NOT T) OR (NOT R AND NOT S AND T)	2

9(a)	1 mark per bullet point, max 2 marks • NOT (A XOR B) • NAND ((B AND C) OR A) X = (NOT (A XOR B)) NAND ((B AND C) OR A) // X = ((B AND C) OR A) NAND (NOT (A XOR B))	2
------	--	----------

Answer 29 (continued)

9(b)	1 mark for 1 or 2 row numbers correct, 2 marks all three correct Row 2 Row 4 Row 7	2
------	---	----------

3(a)	1 mark per shaded area, max 2 marks <table><tr><th>A</th><th>B</th><th>C</th><th></th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C		X	0	0	0		1	0	0	1		0	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		1	2
A	B	C		X																																											
0	0	0		1																																											
0	0	1		0																																											
0	1	0		1																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		1																																											
3(b)	1 mark per bullet point, max 2 marks • A AND B OR C • C NOR D and final two gates 	2																																													

6(a)	1 mark for each correct answer $Y = \text{NOT } (B \text{ OR NOT } C)$ $Z = E \text{ NOR } (\text{NOT } C \text{ XOR } D)$															
6(b)	1 mark for drawing a truth table with correct inputs 1 mark for the correct outputs in the NAND column <table><tr><th>Input 1</th><th>Input 2</th><th>NAND</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	Input 1	Input 2	NAND	0	0	1	0	1	1	1	0	1	1	1	0
Input 1	Input 2	NAND														
0	0	1														
0	1	1														
1	0	1														
1	1	0														

8(a)	1 mark for A column and 1 mark for B column <table><tr><th>X</th><th>Y</th><th>Z</th><th>Working space</th><th>A</th><th>B</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td><td>0</td></tr></table>	X	Y	Z	Working space	A	B	0	0	0		0	0	0	0	1		0	0	0	1	0		0	0	0	1	1		0	1	1	0	0		0	0	1	0	1		0	1	1	1	0		1	0	1	1	1		1	0
X	Y	Z	Working space	A	B																																																		
0	0	0		0	0																																																		
0	0	1		0	0																																																		
0	1	0		0	0																																																		
0	1	1		0	1																																																		
1	0	0		0	0																																																		
1	0	1		0	1																																																		
1	1	0		1	0																																																		
1	1	1		1	0																																																		
8(b)	1 mark for a correct answer • NOT (A XOR B) • NOT A XOR B // NOT B XOR A • NOT (A OR B) OR (A AND B) // (NOT A AND NOT B) OR (A AND B) • (NOT A AND NOT B AND NOT C) OR (NOT A AND NOT B AND C) OR (A AND B AND NOT C) OR (A AND B AND C) • A AND B OR (A NOR B)																																																						

2(a)	1 mark for each correct column <table><tr><th>X</th><th>Y</th><th>Z</th><th>Working space</th><th>A</th><th>B</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td><td>1</td></tr></table>	X	Y	Z	Working space	A	B	0	0	0		1	1	0	0	1		0	0	0	1	0		1	0	0	1	1		1	1	1	0	0		1	1	1	0	1		1	0	1	1	0		1	0	1	1	1		1	1
X	Y	Z	Working space	A	B																																																		
0	0	0		1	1																																																		
0	0	1		0	0																																																		
0	1	0		1	0																																																		
0	1	1		1	1																																																		
1	0	0		1	1																																																		
1	0	1		1	0																																																		
1	1	0		1	0																																																		
1	1	1		1	1																																																		
2(b)	1 mark for two correct NAND and OR																																																						